

Study Report: ServiceNow Developer / Administrator - Career Path & Core Skills

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Family:	ServiceNow Developer / Administrator
Level:	Beginner to Advanced
Tracks:	Beginner, Intermediate, Advanced
Chapters:	18
Source:	Official docs, free libraries, and industry-standard references (see sources and books)

Official sources and free libraries

- <https://developer.servicenow.com/dev.do>
- <https://docs.servicenow.com/>
- <https://learning.servicenow.com/>
- <https://www.servicenow.com/products/itsm.html>
- <https://www.servicenow.com/community/developer-articles/>

https://docs.servicenow.com/bundle/zurich-api-reference/page/script/server-scripting/concept/c_GlideRecordAPI.html

Summary

ServiceNow career path covering ITSM incident, problem, and change workflows, CMDB foundation data, Flow Designer automation, GlideRecord server-side scripting, Service Portal widgets, REST integrations, and scoped application development. Built from ServiceNow Developer documentation, Now Learning, and CSA/CIS certification objectives.

Skill stack by level

Recommended books

1. Learning ServiceNow - Tim Woodruff

Level: Beginner | Platform navigation, incidents, CMDB, and admin fundamentals.

2. ServiceNow Development Handbook - Tim Woodruff

Level: Intermediate | GlideRecord, business rules, script includes, and integrations.

3. ServiceNow Documentation - ServiceNow

Level: Beginner | Official ITSM, Flow Designer, and developer references.

4. ServiceNow Certified System Administrator - ServiceNow Learning

Level: Intermediate | Official CSA exam path and platform configuration skills.

5. ServiceNow IT Operations Management - Packt

Level: Advanced | Event management, discovery, and operational workflows.

End-to-end projects

Project 1: ITSM Incident Management Setup

Beginner | 2-3 weeks - Configure incident table, assignment groups, SLAs, and email-to-incident on Personal Developer Instance.

- Groups: Group structure
- SLA: SLA records active
- Email: Email creates incident demo
- Test: Test incident screenshots

Project 2: Employee Onboarding Flow

Intermediate | 3-4 weeks - Flow Designer automation: new hire triggers tasks for IT, HR, and facilities with approval steps.

- Flow: Flow diagram export
- Tasks: Task templates
- Approval: Approval demo video
- Dashboard: Dashboard screenshot

Project 3: CMDB + Event Management Integration

Advanced | 4-5 weeks - Populate CMDB with server CIs, integrate monitoring alerts to auto-create incidents with enrichment.

- CMDB: CMDB populated
- Discovery: CMDB diagram
- Events: Auto-incident demo
- Impact: Impact analysis report

Chapters

Track: Beginner

Chapter 1: ServiceNow Platform and Instance Navigation [Beginner]

Chapter focus: ServiceNow Platform and Instance Navigation *(Level: Beginner)**

ServiceNow is a cloud platform built on a single data model with tables, records, and roles. Administrators use Application Navigator, lists, forms, and filters daily. Personal Developer Instances (PDIs) let you practice without risking production. Understand global vs scoped applications and the difference between configuration and customization.

Code Reference:

```
gs.info('CodeQuest learner exploring instance: ' + gs.getSession().getClientIP());
```

```
// Application Navigator -> Incident -> Open
```

What it shows: Server-side gs.info logs help trace script execution in System Logs.

Topic guide

What it actually is

The Now Platform is a low-code enterprise workflow and data platform.

When to use it

When organizations standardize ITSM, HR, or customer service on one system of record.

Where to use it

Fortune 500 IT departments, MSPs, universities, and government agencies.

Real use example

A CodeQuest cohort provisions PDIs, favorites the Incident app, and completes the 'Welcome to ServiceNow' guided setup before touching production-like data.

Chapter 2: ITSM Incident Management Fundamentals [Beginner]

Chapter focus: ITSM Incident Management Fundamentals** *(Level: Beginner)

Incidents restore normal service quickly. Core fields: caller, category, priority (impact × urgency), assignment group, and state (New ? In Progress ? Resolved ? Closed). SLAs attach response and resolution clocks. Major incident management coordinates war rooms for widespread outages.

Code Reference:

```
var gr = new GlideRecord('incident');
gr.addQuery('priority', '1');
gr.query();
while (gr.next()) {
    gs.info('P1: ' + gr.number);
}
```

What it shows: GlideRecord queries open P1 incidents for morning standup dashboards.

Topic guide

What it actually is

Incident management is the reactive restore-service ITIL practice on ServiceNow.

When to use it

When users report broken services: VPN down, email bounce, portal 500 errors.

Where to use it

Service desk, NOC, and application support queues.

Real use example

During finals week a learning portal returns 503; P1 incident auto-assigns to Web-Ops with executive notification from a Flow Designer trigger.

Chapter 3: Problem, Change, and Request Fulfillment [Beginner]

Chapter focus: Problem, Change, and Request Fulfillment *(Level: Beginner)**

Problems investigate root cause behind recurring incidents. Changes control risk through CAB approval, schedules, and backout plans. Service Catalog and Request items fulfill standard offerings like laptop requests. Link records: incident ? problem ? change ? knowledge article.

Code Reference:

```
// Change state workflow: New -> Assess -> Authorize -> Scheduled -> Implement -> Review
// Request: sc_req_item table
```

What it shows: ITIL practice tables interconnect for traceable service lifecycle.

Topic guide

What it actually is

These practices move teams from firefighting to controlled improvement.

When to use it

Mature IT organizations with CAB meetings and known error databases.

Where to use it

Enterprise change windows, hardware refresh programs, and software rollouts.

Real use example

Repeated Moodle login failures spawn a problem record; authorized change adds SSO patch Saturday 2 AM.

Chapter 4: Users, Groups, Roles, and ACLs [Beginner]

Chapter focus: Users, Groups, Roles, and ACLs *(Level: Beginner)**

Security in ServiceNow is role-based. Groups bundle users; roles grant table and field access. ACLs (Access Control Lists) evaluate create/read/update/delete row and field conditions. Always apply least privilege: grant itil role to agents, not admin, unless justified.

Code Reference:

```
// ACL example condition script:
// answer = current.assigned_to == gs.getUserID();
```

What it shows: ACL condition scripts restrict records to assignee or group members.

Topic guide

What it actually is

ACLs enforce who sees which records at the database layer.

When to use it

Every table with sensitive HR, security, or financial data.

Where to use it

Multi-department instances sharing incident data but isolating salary tables.

Real use example

Student workers get scoped read on their own incidents via ACL and cannot view executive tickets.

Chapter 5: Lists, Filters, and Reporting Basics [Beginner]

Chapter focus: Lists, Filters, and Reporting Basics *(Level: Beginner)**

Lists display query results; filters save conditions for reuse and dashboards. Reporting supports bar, pie, trend, and heat maps. ServiceNow Performance Analytics adds indicators and breakdowns. Export responsibly-PII in CSV exports violates policy.

Code Reference:

```
var ga = new GlideAggregate('incident');
ga.addAggregate('COUNT');
ga.groupBy('category');
ga.query();
while (ga.next()) {
    gs.info(ga.category + ': ' + ga.getAggregate('COUNT'));
}
```

What it shows: GlideAggregate counts incidents by category for weekly management reports.

Topic guide**What it actually is**

Lists and reports turn ticket data into operational insight.

When to use it

Weekly service review meetings and SLA compliance tracking.

Where to use it

Manager dashboards, team standups, and executive QBR decks.

Real use example

CodeQuest mentor reviews category breakdown showing 35% password-reset tickets-drives KB investment.

Chapter 6: Service Portal Introduction [Beginner]

Chapter focus: Service Portal Introduction *(Level: Beginner)**

Service Portal delivers branded self-service with pages, widgets, and themes. Employees search knowledge, submit requests, and track incidents without calling the desk. Widgets use AngularJS client controllers and server scripts. UI Builder is the modern successor for Experience workflows.

Code Reference:

```
// Client controller snippet
function($scope, spUtil) {
  $scope.submit = function() {
    spUtil.update('incident');
  };
}
```

What it shows: Client controllers bind form actions to Service Portal record producers.

Topic guide**What it actually is**

Service Portal is the employee-facing front door to ITSM.

When to use it

When deflecting Tier 1 tickets through self-service and knowledge.

Where to use it

Corporate intranets, university IT portals, and HR service centers.

Real use example

Faculty portal adds 'Report Wi-Fi Issue' widget; submissions create pre-categorized incidents.

Track: Intermediate

Chapter 7: CMDB and Configuration Items [Intermediate]**Chapter focus: CMDB and Configuration Items** *(Level: Intermediate)**

The CMDB stores Configuration Items (CIs)-servers, databases, applications-with attributes and relationships. Accurate CI data powers change impact analysis and incident assignment. Identification rules reconcile duplicates; baselines track drift. Avoid garbage-in by governing manual CI creation.

Code Reference:

```
var rel = new GlideRecord('cmdb_rel_ci');
rel.addQuery('parent', serverSysId);
rel.query();
while (rel.next()) {
  gs.info('Depends on: ' + rel.child.name);
}
```

What it shows: Relationship queries reveal downstream CIs before server patching.

Topic guide**What it actually is**

CMDB is the technical inventory graph underpinning IT operations.

When to use it

Change planning, outage communication, and root cause mapping.

Where to use it

Data centers, hybrid cloud estates, and application portfolios.

Real use example

Planned DB maintenance queries CMDB to notify all apps depending on SQL-PROD-03.

Chapter 8: Flow Designer vs Workflow Editor [Intermediate]

Chapter focus: Flow Designer vs Workflow Editor *(Level: Intermediate)**

Flow Designer is the modern, low-code automation tool with triggers, conditions, and actions across tables. Legacy Workflow Editor still exists on older instances-know both. Flows can call subflows, ask for approval, invoke scripts, and integrate via spokes. Test in subprod with rollback plans before production promotion.

Code Reference:

```
// Flow trigger: Record Created on incident
// Action: If priority == 1 -> Send Email + Create Major Incident task
```

What it shows: Declarative flows replace hundreds of lines of custom workflow code.

Topic guide**What it actually is**

Flow Designer automates multi-step business processes with observability.

When to use it

SLA breaches, approvals, notifications, and cross-table updates.

Where to use it

HR onboarding, access provisioning, and major incident orchestration.

Real use example

New-hire flow creates AD request, hardware catalog item, and Slack notification automatically.

Chapter 9: GlideRecord Server-Side Scripting [Intermediate]

Chapter focus: GlideRecord Server-Side Scripting *(Level: Intermediate)**

GlideRecord is the primary API for querying and mutating records server-side. Use addQuery, setLimit, chooseFields, and getValue/setValue. Avoid GlideRecord in client scripts-use GlideAjax instead. Performance matters: nested loops on large tables cause transaction timeouts.

Code Reference:

```
var gr = new GlideRecord('incident');
gr.addEncodedQuery('active=true^priorityIN1,2');
gr.orderBy('opened_at');
```

```
gr.setLimit(100);
gr.query();
```

What it shows: Encoded queries mirror list filter syntax for maintainable scripts.

Topic guide

What it actually is

GlideRecord is the server-side database access layer.

When to use it

Business rules, scheduled jobs, script includes, and fix scripts.

Where to use it

Any automation touching multiple records or complex conditions.

Real use example

Scheduled job reopens stale P2 incidents lacking updates after 48 hours for manager review.

Chapter 10: Business Rules, Client Scripts, and UI Policies [Intermediate]

Chapter focus: Business Rules, Client Scripts, and UI Policies** *(Level: Intermediate)

Business rules run server-side on insert/update/delete (before or after). Client scripts react in the browser onLoad, onChange, onSubmit. UI policies show/hide mandatory fields without code when possible. Order of execution matters: UI Policy ? Client Script ? Business Rule ? Flow.

Code Reference:

```
// Business Rule (before): if (current.priority == 1) current.assignment_group = 'Major Incident';
```

What it shows: Before business rules mutate records prior to database commit.

Topic guide

What it actually is

These artifacts implement field logic and data integrity at multiple tiers.

When to use it

Form behavior, validation, and auto-population scenarios.

Where to use it

Incident categorization, change risk scoring, and catalog variable logic.

Real use example

Changing category to 'Network' auto-assigns Network-Ops via onChange client script and business rule.

Chapter 11: Service Catalog and Record Producers [Intermediate]

Chapter focus: Service Catalog and Record Producers** *(Level: Intermediate)

Service Catalog presents orderable items with variables, workflows, and fulfillment tasks. Record producers create records (often incidents) from portal forms. Variable sets reuse common questions. Price and recurring price fields support chargeback reporting.

Code Reference:

```
// Catalog item variables: u_device_type, u_justification
// Produces: sc_req_item -> requested item tasks
```

What it shows: Catalog variables map to task and CI fields downstream.

Topic guide

What it actually is

Service Catalog standardizes fulfillment for repeated requests.

When to use it

Hardware refresh, software licenses, access grants, and facilities.

Where to use it

Employee self-service and CodeQuest sandbox 'request sandbox VM' lab item.

Real use example

Developer requests cloud sandbox; catalog flow creates approval, provisioning task, and closure survey.

Chapter 12: Integration Hub and REST Messages [Intermediate]

Chapter focus: Integration Hub and REST Messages *(Level: Intermediate)**

Integration Hub provides spokes and connections to external systems (Jira, Azure, Slack). REST Message defines endpoints, headers, and authentication. MID Server reaches on-prem targets. Log integration errors to dedicated tables and build retry subflows for resilience.

Code Reference:

```
var rm = new sn_ws.RESTMessageV2('CodeQuest_Asset_API', 'get_assets');
var response = rm.execute();
var body = response.getBody();
```

What it shows: RESTMessageV2 calls external asset APIs from a scheduled data pull.

Topic guide

What it actually is

Integrations connect ServiceNow to the wider enterprise toolchain.

When to use it

Bi-directional ticket sync, CMDB enrichment, and chatops alerts.

Where to use it

Multi-tool IT stacks where ServiceNow remains the system of engagement.

Real use example

Jira defect linked to ServiceNow incident via Integration Hub spoke keeps dev and ops aligned.

Chapter 13: Update Sets and Deployment Pipeline [Intermediate]

Chapter focus: Update Sets and Deployment Pipeline** *(Level: Intermediate)

Update Sets capture configuration changes for promotion across instances (dev ? test ? prod). Scoped apps use source control and pipelines in modern shops. Preview conflicts before merge. Never edit prod directly-route through change control with peer review.

Code Reference:

```
// Compare Update Sets: Incidents_Flow_v2
// Pipeline: Dev PDI -> Team Dev -> QA -> Prod (Change CHG001234)
```

What it shows: Promotion path mirrors software delivery lifecycle.

Topic guide

What it actually is

Controlled promotion prevents untested changes in production.

When to use it

Every enterprise with segregated ServiceNow instances.

Where to use it

Agile product teams shipping weekly catalog improvements.

Real use example

Flow fix tested in QA update set; CAB-approved change deploys Saturday with rollback set ready.

Track: Advanced

Chapter 14: Scoped Application Development [Advanced]

Chapter focus: Scoped Application Development** *(Level: Advanced)

Scoped applications isolate code and data with application-specific tables and ACLs. Use ServiceNow Studio, application files, and ATF tests. Script Includes encapsulate reusable logic. Follow ServiceNow JavaScript APIs-avoid unsupported DOM hacks.

Code Reference:

```
var AssetUtils = Class.create();
AssetUtils.prototype = Object.extend(Object(global.AbstractAjaxProcessor, {
  process: function() { return this.getParameter('sysparm_query'); }
}));
```

What it shows: Script Includes extend AbstractAjaxProcessor for GlideAjax endpoints.

Topic guide

What it actually is

Scoped apps package maintainable custom solutions on the platform.

When to use it

ISVs, internal platform teams, and complex custom processes.

Where to use it

Custom vendor management app with own tables, modules, and portal.

Real use example

CodeQuest capstone ships scoped inventory app through store-ready checklist and ATF suite.

Chapter 15: Performance and GlideAggregate Optimization [Advanced]

Chapter focus: Performance and GlideAggregate Optimization** *(Level: Advanced)

Slow queries hurt UX and batch jobs. Use indexes, avoid dot-walking in large loops, prefer GlideAggregate to manual counting. Enable Debug Business Rule and SQL explain plans in non-prod. Partition long-running jobs with setLimit and resume markers.

Code Reference:

```
var ga = new GlideAggregate('task');
ga.addAggregate('COUNT');
ga.addQuery('assignment_group', groupId);
ga.setGroup(false);
ga.query();
var count = ga.next() ? ga.getAggregate('COUNT') : 0;
```

What it shows: Aggregate queries replace thousands of getRowCount loops.

Topic guide

What it actually is

Performance tuning keeps instances responsive at enterprise scale.

When to use it

Instances exceeding millions of task rows or heavy integrations.

Where to use it

Global enterprises and high-volume MSP instances.

Real use example

Dashboard widget refactored from GlideRecord loop to GlideAggregate-load time drops 18s to 0.4s.

Chapter 16: Scripted REST APIs and OAuth [Advanced]

Chapter focus: Scripted REST APIs and OAuth** *(Level: Advanced)

Scripted REST APIs expose custom endpoints with scripted processing and ACL enforcement. Define resources, HTTP verbs, and API security profiles. OAuth 2.0 and JWT bearer tokens gate access for mobile apps and external dashboards.

Code Reference:

```
(function process(/*RESTAPIRequest*/ request, /*RESTAPIResponse*/ response) {
    var body = request.body.dataString;
    response.setStatus(200);
    response.setBody({status: 'ok'});
})(request, response);
```

What it shows: Scripted REST handlers validate payloads and return JSON responses.

Topic guide

What it actually is

Scripted REST enables platform-as-backend for custom experiences.

When to use it

Mobile apps, partner portals, and machine-to-machine automation.

Where to use it

Secure external integrations without sharing user credentials.

Real use example

CodeQuest mobile app fetches open incidents via OAuth-secured Scripted REST for student interns.

Chapter 17: Domain Separation and Enterprise Scale [Advanced]

Chapter focus: Domain Separation and Enterprise Scale** *(Level: Advanced)

Domain separation partitions data for MSPs or conglomerates. Domain visibility rules control cross-domain access. Process separation duplicates tables per domain-heavyweight. Design domains early; retrofits are painful.

Code Reference:

```
gs.getSession().setDomainID(domainSysId);
var gr = new GlideRecord('incident');
gr.query(); // returns only domain-visible records
```

What it shows: Session domain context filters queries automatically.

Topic guide

What it actually is

Domain separation multi-tenants one instance safely.

When to use it

MSPs managing hundreds of customers on shared infrastructure.

Where to use it

Global companies with regional IT autonomy.

Real use example

MSP adds new school district domain; agents see only that district's incidents and CIs.

Chapter 18: Event Management and ITOM [Advanced]**Chapter focus: Event Management and ITOM** *(Level: Advanced)**

Event Management correlates monitoring alerts into actionable incidents. Connectors ingest from SolarWinds, Datadog, or Nagios. MOGs group related events; CI binding ties alerts to CMDB. Threshold rules reduce noise before ticket spam.

Code Reference:

```
// Event rule: 3+ CPU alerts on same CI in 10 min -> create incident P2
```

What it shows: Correlation rules prevent alert storms from flooding the desk.

Topic guide**What it actually is**

ITOM practices bridge monitoring tools and ServiceNow workflow.

When to use it

NOC environments with thousands of daily monitoring events.

Where to use it

Hybrid cloud shops with tool sprawl.

Real use example

Database failover generates 50 alerts; correlation opens one major incident with linked events.